# SAINT MICHAEL COLLEGE OF CARAGA

**SOURCE ATTRIBUTION OF MICROPLASTICS IN SEAWATER**

A Thesis Proposal To
The Faculty of
College of Teacher Education

In Partial Requirements for the Degree
Bachelor of Secondary Education Major in Science

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2. **List and Explanation**
•Example: *"To my parents, for their love and support, and to my advisor, Dr. [Name], for invaluable guidance."*
3.**Closing (Optional)**
•Example: *"This work would not have been possible without you."*

## ACKNOWLEDGEMENTS

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**Suggested Structure**

**Opening Statement**
• Express general gratitude for the opportunity to complete the research or thesis.

**Acknowledgment of Advisors and Professors**
• Thank your thesis advisor, committee members, or professors for their guidance and expertise.

**Acknowledgment of Family and Friends**
•Thank those who provided emotional or financial support.

**Acknowledgment of Institutional or Financial Support**
•Recognize scholarships, grants, or organizations that funded or facilitated your research.

**Special Thanks (Optional)**
Mention individuals who provided significant help, such as peers, mentors, or technical staff.

**Closing Statement**
•Conclude with a general expression of gratitude.

**Example**

*I am deeply grateful to everyone who contributed to the completion of this thesis. My sincerest thanks to my advisor, Dr. [Name], for your invaluable insights and guidance. To my committee members, thank you for your constructive feedback and support.*

*To my family, your love and encouragement kept me going through this journey, and to my friends, your support meant the world to me. I would also like to thank [Institution/Organization] for funding this research.*

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**Include all tables in your paper. Make sure that the tables are properly accounted for.**

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# CHAPTER 1

# THE PROBLEM AND ITS SCOPE

## 1.1 Background of the study

Plastics are currently too abundant and simply unstoppable (Castellón-Mena et al., 2024). This makes the Philippines generate 2.7 million tons of plastic waste annually into the ocean (Meijer et al., 2021); (World Bank Group, 2021), a consequence of humans increasing reliance on these cheap, long-lasting, and lightweight materials (S. George (2020); (Pilapitiya & Ratnayake, 2024). This dependence, however, has led to widespread environmental concerns (Paterson, 2019), with macroplastics (larger than 5 mm) now prevalent in both aquatic and terrestrial ecosystems (Pacilan & Bacosa, 2022). Primarily source from human activities (Retnaningdyah1 & Vidayant, 2025), this pollution poses various environmental risks, including entanglement of marine life, suffocation of mangrove root systems (M. George et al., 2023), and the release of toxic compounds and microplastics into the seas and food chains (Neelavannan et al., 2025). Identifying the sources and pathways of this degradation is important for creating effective strategies to mitigate marine plastic pollution. The researchers combat plastic pollution to achieve Sustainable Development Goals (SDGs) related to health (SDG 3), sustainable cities (SDG 11), consumption patterns (SDG 12), and marine life (SDG 14) by implementing waste management (John N. Hahladakis, 2025), recycling, and promoting sustainable consumption (Muñiz & Rahman, 2025). However, despite international measures, considerable amounts of plastic waste remain mixed with municipal waste, prohibiting recycling and recovery (Todorova et al., 2025).

From the northern to the southern Bulgarian Black Sea coast, plastic pollution is a global issue as plastics enter the sea through various pathways, including rivers, fishing activities, urban runoff, recreational activities, and natural disasters. Once in the ocean, plastics move across borders due to wind, currents, and tides, and often accumulate on beaches, where they can

degrade and eventually return to the ocean (Simeonova & Chuturkova, 2020). However, the focus on the spatial distribution of single-use plastics and fishing gear was limited, and there is a need to identify more plastic distribution in the ocean. According to Zambrano-Pinto et al. (2024), human activity is the reason for the plastic pollution found in beach sediments from the Urias coastal lagoon in northwest Mexico. Their results highlighted the size and attributes of plastic pollution in this ecosystem by revealing a prominent presence of macroplastics. The study highlights the importance of better waste management techniques to protect the economic and ecological functions of the lagoon. Moreover, the Global South, particularly Southeast Asia, is unequally impacted by the plastic pollution issue.

In Sultan Naga Dimapor, the longest coastline in the province of Lanao del Norte and a high-coastal community, many people raise concerns about the extent of plastic pollution and focus on the amount of plastic that ends up on the seabed. To determine the prevalence, types, and density of macroplastic litter on the coastal seabed (Pacilan & Bacosa, 2022), the researchers from Sultan Naga Dimapor collected and analyzed plastic litter from the municipality to establish a baseline for macroplastic litter in the Philippines. In Pujada Bay, Davao Oriental, anthropogenic marine debris and oceanic solid waste significantly affect marine biodiversity and ecosystems, especially in mangrove stands. The researchers in the study by Abreo et al. (2021) collected debris and then compared weights in different material categories within each area. These researchers used statistical analyses to determine differences in plastics (Abreo et al., 2021; Lumasag et al., 2024). However, neither study provides information on marine debris or plastic sources. Moreover, in Barangay Camagong seawater in Nasipit, Agusan del Norte, bioaccumulation of microplastics in oysters was studied through sample collection and FTIR spectroscopy; however, the sources of microplastics were not included in Lumasag et al. (2024).

In Barangay Camagong, Nasipit, Agusan del Norte, the increasing entry of plastic trash is a pressing ecological and social issue. The improper disposal of domestic and commercial waste has led to continuous plastic pollution in coastal waters, affecting aquatic organisms through plastic degradation (Kibria et al., 2023). From the study of Lumasag et al. (2024), a residential site in Barangay Camagong, Nasipit, Agusan del Norte, indicated the presence of microplastics, specifically D polyvinyl stearate and PVC with Citric Ester in oyster flesh. This finding underscores the urgent need to identify and quantify the sources of microplastic contamination in the seawater. Working on a single site, such as Barangay Camagong, offers benefits for experimental research. Studies have shown that sites with human waste are often sources of aquatic pollution (Kye et al., 2023). Plastics enter the ocean through various pathways, including land-based sources from domestic waste and sea-based sources from fishing activities and maritime shipping (Thushari & Senevirathna, 2020). Mapping and classifying microplastic sources, especially in Barangay Camagong will provide significant information on the sources of microplastics and the migration of macroplastics within the barangay's marine ecosystem (Pierard et al., 2022).

The results of this study will benefit the stakeholders residing in Barangay Camagong, Nasipit, Agusan Del Norte, by providing evidence of macroplastics and their dispersal of microplastics, which is essential to understand the source attribution of seawater microplastics and to determine the origin of the pollutants and their transport in the marine environment (Ashrafy et al., 2023; Pierard et al., 2022). The Municipal Environment and Natural Resources Office (MENRO) and the Local Government Unit of Nasipit can use the findings to enhance coastal cleanup drives and improve proper community waste disposal, thereby reducing environmental dumping throughout the municipality. Saint Michael College of Caraga hopes to apply the results of this research to strengthen and innovate its current waste management programs, encouraging

sustainable practices within the institution and communities. The results of this study are to provide detailed, evidence-based information that will help residents, policymakers, and other stakeholders develop more effective solid waste management strategies, by that means supporting better environmental quality and public health gains.

**1.2 Review of Related Literature**

This chapter will provide the foundational background for the research, incorporating ideas and support from various scholarly works. The literature review aims to study macroplastic pollution in seawater in Barangay Camagong, Nasipit, Agusan del Norte and address the significant reasons and factors that contribute to plastic accumulation, the description of macroplastic types, sizes, sources, and polymer composition, contamination of coastal waters by land-based and ocean-based plastics, the possible environmental and ecological risks of macroplastic pollution, and the monitoring and analytical methods to assess plastic pollution. The findings will inform key stakeholders, including the Local Government Unit of Nasipit, the Municipal Environment and Natural Resources Office of Nasipit, the Department of Environment and Natural Resources- City Environment and Natural Resources Office (DENR-CENRO) of Nasipit, and the Barangay Captain of Barangay Camagong, Nasipit, Agusan del Norte. The researchers will present the findings to establish connections between various theories and research outcomes.

**Literature of the Study**

**Macroplastic**

Macroplastics are becoming an environmental concern due to their presence in ecosystems and the damage they cause to marine, freshwater, and even terrestrial environments (Al-Zawaidah et al., 2021). Macroplastics have many sources, including daily single-use products like bags, bottles, cups, and straws, as well as fishing litter such as nets and lines (Bimali Koongolla

et al., 2018). Single-use plastics, especially, are a significant issue; they pollute water heavily, often because they are thrown around recklessly or come from domestic waste (Blettler & Mitchell, 2021).

Over time, environmental factors like UV light, mechanical stress, and microbial breakdown break these larger plastics down into smaller pieces, ultimately forming microplastics —particles smaller than 5 mm. This process of transformation has been well established in research on the degradation of plastics in aquatic ecosystems (Alimba & Faggio, 2019). The breakdown of macroplastics into microplastics increases pollution and complicates mitigation efforts, as smaller particles are more challenging to detect and remove from ecosystems.

**Categories of Macroplastics**

Plastic waste is an issue in the world's oceans and waterways (The National Research of the Philippines, n.d.). Individuals use plastic items primarily for basic needs, such as food security, transportation, housing, and sanitation (Uknowledge & Meyer, 2021). Poor waste management and improper disposal of plastic materials result in their accumulation as waste, much of which ends up in the ocean, causing pollution (Abalansa et al., 2020). Plastic waste categories that find their way into the sea are (a) food packaging, (b) disposable utensils, (c) food containers (d) cloth, (e) plastic caps, (f) sanitary napkin and diapers, (g) ropes and nylon, (h) cigarette butt, (i) plastic fragments, (j) sando bags, (k) styrofoam, (l) medicinal packaging, (m) sack, (n) plastic bottle, (o) cellophane and (p) other plastics (Pacilan & Bacosa, 2022; Sajorne et al., 2021). In the Adriatic Sea region, degrades or fragments of plastics happen over time, resulting in an abundance of microplastic debris (Gomiero et al., 2019). The sampling segregation was based on what types of plastics occur in terms of size, polymer type, and shape (Gomiero et al., 2019; Jemaa et al., 2021; Maharana et al., 2020). A study by Râpă et al. (2024) revealed that, among the marine plastic litter

collected, single-use plastics (SUPs), such as bottles, bags, food wrappers, and packaging materials, were the most frequent items. Also, under the Atlantic Ocean, broken plastics, such as intact bottles, bags, and nets, are the dominant type of plastic waste. In the mangrove areas around Cebu Island, Philippines, fragments like plastic bags and packaging items are the most common types of plastic litter originating from nearby communities and fishing activities (Paler et al., 2022). The National Oceanic and Atmospheric Administration concluded that the commonly found plastics in the ocean include cigarette butts, food wrappers, beverage bottles, straws, cups and plates, bottle caps, and single-use bags.

**Microplastics**

Given that a significant proportion of packaging is designed for single use, it frequently accumulates in the environment (Tramoy et al., 2025). The degradation of these larger plastic items results in smaller particles, commonly referred to as microplastics. Research conducted in Laguna de Bay indicates a positive correlation between microplastic concentration and population density. In this ecosystem, the majority of microplastics are fibrous in nature (57%), with a predominance of blue coloration (53%). The most prevalent polymers identified include polypropylene (PP), ethylene vinyl acetate copolymer (EVA), and polyethylene terephthalate (PET), collectively accounting for 65% of the total microplastic composition (Arcadio et al., 2022). Furthermore, a study performed in Binh Dinh, Vietnam, analyzed microplastic content in the digestive systems of various fish species, categorizing the particles by type, size, and color. Other marine organisms, specifically mollusks like oysters, are identified as indicators of microplastic pollution due to their filter-feeding habits (Villaceran et al., n.d.; Wootton et al., 2022). Microplastics were detected in all specimens, with microfibers being the most abundant type, and polyethylene, polyvinyl ether, and polymethacrylate identified as primary constituents. A research

gap remains in the comprehensive understanding of the long-term ecological and human health implications associated with this microplastic contamination (Tran et al., 2023). While microplastics are recognized as complex, multifaceted pollutants, the precise source of specific plastics of their accumulation within the tissues and organs of marine organisms requires further qualitative and quantitative investigation.

**Polymer Types of Plastics**

According to the research conducted by Pacilan & Bacosa (2022) and Sajorne et al. (2021), plastic polymer categories are polyethylene terephthalate (PET), high-density polyethylene (HDPE), polyvinyl chloride (PVC), low-density polyethylene (LDPE), polypropylene (PP), polystyrene (PS), polyamide (PA), multilayer, and others. The results of the study indicate that knowing the types of polymers traces the origins of pollution and identifies the sources of the plastics, whether from residential areas, mangroves, beaches, or non-residential locations in the beaches of Puerto Princesa, Palawan Island, and in the coastal sea beds of Sultan Naga Dimaporo, Lanao Del Norte, Philippines. In Myanmar's Ayeyarwady coastal regions, the research also found that the two most prevalent plastic polymers in marine litter were polypropylene (PP) and polyethylene (PE). Such materials are associated with consumer products such as food and beverage packaging (LYNN et al., 2025). Different polymers also have distinct properties that affect buoyancy, rates of degradation, and interactions with seas. For instance, polyethylene (PE) and polypropylene (PP) float and may drift long distances, whereas polyvinyl chloride (PVC) sinks and settles in sediments (Pakhomova et al., 2020). It is crucial to determine polymer types from plastic because various polymers have varied densities and degradation rates. This determines whether they float, sink, or disintegrate, hence having an influence on their distribution in marine environments (Dumon, 2025).

**Coastal Population Density**

Coastal population density understands the pressures placed on the marine environment (Reimann et al., 2023). Most beaches in the Philippines were classified as filthy to extremely bad by the Clean-Coast Index (CCI) in several studies on plastic pollution monitoring (Sajorne et al., 2021). Many studies have identified a positive relationship between human population density and macroplastic abundance. Areas with higher coastal populations generate more plastic waste, which enters the marine environment through poor waste management, sewage, and surface runoff (Peng et al., 2020). Recent regional assessments confirm that densely populated shorelines consistently report higher macroplastic loads than sparsely inhabited areas (Acot et al., 2022).

Higher coastal population density increases the likelihood that mismanaged waste will enter rivers and coastal waters, thereby leading to greater macroplastic accumulation along shorelines (Pinto et al., 2024). Over time, these macroplastic fragments, caused by solar radiation, mechanical abrasion, and wave action, produce secondary microplastics that persist in marine ecosystems (Thushari & Senevirathna, 2020). Thus, in densely populated coastal areas such as Barangay Camagong, Nasipit, Agusan del Norte, poor waste management, coupled with increasing human activity, can significantly elevate macroplastic deposition, which may later fragment into microplastics, affecting marine life (Marmara et al., 2023). While substantial progress has been made in quantifying coastal population density and macroplastic pollution, gaps remain in integrating these variables. Many studies focus on demographic analysis or pollution monitoring, but few explicitly link standardized population measures to plastic abundance in a comparative framework (Liu et al., 2022).

**Environmental Factors**

Large plastic debris, such as macroplastics, can cause entanglement, ingestion, or blockage of digestive tracts in marine animals, including fish, sea turtles, seabirds, and marine mammals (Marmara et al., 2023). Macroplastics act not only as direct hazards but also as pioneers of microplastics, thereby indirectly increasing the burden of plastic pollution in marine ecosystems (Bacha et al., 2023). Even if ingestion does not lead to immediate death, it can produce sublethal, chronic effects on the organisms (Lechthaler et al., 2020).

Marine animals are known to mistake floating macroplastics for prey or become entangled in larger plastic items. The presence of macroplastics contributed by rivers adds to the volume of items in the marine environment that can entangle or be ingested by fish, marine mammals, seabirds, turtles, etc. (Al-Zawaidah et al., 2021). Macroplastics remain the primary contributors to plastic pollution, both as physical pollutants and as the source materials that produce microplastics. The environmental consequences of macroplastic pollution in marine environments are wide-ranging, affecting individual organisms, habitats, and ecosystem functions (Inocente et al., 2025). These macroplastics act as mobile pollutants, entering rivers from earthly sources and distributed along river channels, banks, and floodplains (Dalu et al., 2025). These interactions near water bodies indicate that macroplastics are redistributed along river corridors, becoming hazards to wildlife in those zones (Blettler & Mitchell, 2021).

According to Simeonova & Chuturkova (2020), household plastics, including food wrappers, bottles, and bags, are major contributors to terrestrial and marine pollution. Improper disposal, such as littering and inefficient waste management, exacerbates this issue, allowing plastics to enter rivers, beaches, and oceans. Industrial activities also contribute to macroplastic pollution in the marine environment and coastal areas, particularly through plastic packaging and

disposable materials used in manufacturing and distribution processes, including packaging from industries like food production, medicine, and chemicals, which have a link to plastic debris found in marine environments (Kumar et al., 2021).

**Seawater**

Seawater accounts for more than 70% of the Earth's surface area and is crucial in providing sustenance for marine life as well as human activities (PLASTIC POLICIES IN THE PHILIPPINES Country Profile 2, 2025). As a result, fisheries, transportation, recreation, and industry are closely linked to economic and social development (Koelmans et al., 2017). However, in recent years, pollution from sources like factories, sewage, and plastic waste has threatened seawater quality (Li et al., 2023). The pollutants consequently lead to the accumulation of plastics and toxic substances, harming marine food webs and lowering water quality (Muñiz & Rahman, 2025). For instance, massive plastic objects in the ocean can destroy habitats, trap marine species, and distribute invasive species (Galgani et al., 2021). Moreover, contaminants on plastic items can enter the food web when humans consume seafood, potentially affecting human health (de Carvalho-Souza et al., 2018). Thus, to safeguard marine environments and protect coastal communities from hazards, it is imperative to manage and monitor seawater quality (Dalu et al., 2025).

**Seabed**

According to Zhu et al. (2024), researchers introduced a global estimate of macroplastic pollution on the ocean floor, highlighting it as a significant source, accounting for 3-11 million metric tons. Macroplastic accumulation is also concentrated in continental margins and high-density areas, where low degradation rates and high impacts on marine life are the norm. Based on the research by Morales Jiménez et al. (2023), macroplastic accumulation on reef substrates

depends on hydrodynamic conditions, object shape, polymer type, and the extent of dirt. Onshore pollution sources, such as industrial discharges, plastic waste, and human waste, particularly affect the ocean bottom ecosystem, highlighting the importance of enhanced marine waste management procedures and increased environmental consciousness (Putera et al., 2024).

**Hyclin Plus**

Hyclin Plus and Milli-Q distilled water play key roles in post-collection processing for macroplastic samples from seawater, ensuring contamination-free source attribution. These reagents are used similarly to microplastics protocols, but scaled for larger debris, such as 5 mm (Rasmussen et al., 2021). In the sample preparation or cleaning step, the plastic pieces were washed to remove surface impurities using a neutral chemical detergent called Hyclin Plus (Hycel brand), then rinsed with tap water, and finally distilled (Milli-Q) water. By eliminating organic pollutants and environmental residues from the plastic surface, neutral detergents like Hyclin Plus increase the precision of spectroscopic or other laboratory characterization (Villegas-Camacho et al., 2024a). Moreover, hyclin plus cleans collection nets, sieves, and storage containers, leveraging its neutral pH to avoid degrading plastic polymers during analysis preparation (Nayebi et al., 2023).

**Milli-Q Distilled Water**

Milli-Q distilled water serves as the gold standard for ultrapure rinsing in plastics studies, minimizing airborne and equipment-derived particle contamination during sample processing (Negrete Velasco et al., 2023). Milli-Q water reduces background contamination during rinsing or blanks because it contains significantly fewer particles than tap water (Jones et al., 2024). To avoid cross-contamination from experimental materials, studies typically use washing techniques, such as thorough washing with distilled water, Milli-Q water, or pre-filtered water (Gao et al., 2023). Milli-Q grade water is ultra-pure, produced by a Milli-Q purification system, and essential for

sensitive analytical techniques such as FTIR. It prevents contaminants from interfering with sample analysis, ensuring accurate results. Additionally, it is crucial to remember that, to avoid cross-contamination, the samples were handled with gloves (Villegas-Camacho et al., 2024a).

**Industrial Discharge**

Industrial discharges contribute heavily to elevated levels of macroplastics in seawater. Most coastal businesses discharge wastewater comprising plastic residues, packaging, and synthetic fibers into the sea without treatment (Koelmans et al., 2017). Regions with high numbers of factories and poor waste management experience particularly significant amounts of macroplastics from manufacturing, fisheries, and shipping (Requiron & Bacosa, 2022). Such pollution harms marine life, lowers water quality, reduces fish populations, and harms tourism (Zhang et al., 2023). Weak regulations and poor enforcement worsen the problem, underscoring the need for improved production practices and effective wastewater treatment to reduce macroplastic pollution in the ocean (Dalu et al., 2025).

**Health Risk**

Macroplastics remain present in seawater, with increasing health impacts on marine life and humans (Koelmans et al., 2017). During decomposition, large plastic waste produces smaller pieces that can absorb toxic chemicals, including heavy metals, persistent organic pollutants (POPs), and pathogens, from water (Lakhiar et al., 2024). When sea creatures consume these plastics, toxins are absorbed into their systems and can enter the human food chain through seafood (de Carvalho-Souza et al., 2018). Chronic exposure to such pollutants may result in endocrine disruption, leading to abnormal menstrual cycles, infertility, weight gain, mood changes, premature puberty, and hormonal imbalances, such as decreased testosterone or estrogen (Lam et al., 2020). Liver injury may present with abdominal pain or swelling, icterus,

vomiting, dark-colored urine, tiredness, and loss of appetite due to the buildup of harmful compounds, including heavy metals and persistent organic pollutants (Sobhani et al., 2022). Neurological conditions resulting from microplastic exposure include headaches, memory loss, sleep disturbance, dizziness, and mood or behavioral changes due to toxins that affect brain activity (Prüst et al., 2020); Rahman et al., 2023). In the meantime, gastrointestinal problems often arise when organisms consume microplastics, leading to inflammation or interference with the gastrointestinal tract, causing stomach pain, bloating, nausea, constipation, and diarrhea, and altering the gut microbiota (Jiang, 2018; Stephanie L Wright, 2017). All these facts prove that there is an urgent need to enhance pollution control and raise people's awareness to decrease human exposure to plastic toxins (Winton et al., 2020).

**Literature of the Method**

**Sample Collection Timing**

The recommended days for collecting macroplastic samples are spread over three days, with sampling conducted on three weekends to capture variations in plastic deposition influenced by human activities and tidal changes (Pacilan & Bacosa, 2022). The weeks' duration is important to determine the clean coast index across multiple days or weeks for temporal trends in a specific site within the study area (Acot et al., 2022; Pacilan & Bacosa, 2022). Sampling should be done approximately one hour after high tide due to its high tidal range (Hanke & Ruiz-Orejón, n.d.; Tackling Marine Litter in the Atlantic Area Analysis of Strategies for the Monitoring and Evaluation of Accumulations of Marine Litter on the Coast Action 5.2: Monitoring the Presence of ML in the Marine Environment WP 5: Monitoring and Data Management, n.d.). Since it is hotter in the morning than in the afternoon, which is already impacted by increased human activity, it is

recommended to get samples in the morning. Moreover, in the afternoon, the sand might be drier than in the morning (Guidelines for Sampling Microplastics on Sandy Beaches, n.d.). Sunny or dry weather reduces such variability, making it preferable for collecting macroplastic samples (Alvarado-Zambrano et al., 2024). During low tide, and to determine the maximum width of the shoreline, sampling should begin within an hour of the low tide and continue for up to three hours after the actual low tide (Onda & Florence, n.d.). From the study of Acot et al. (2022), the collection of macroplastics was conducted on Wednesdays to represent the weekdays and Saturdays to represent the weekends. It is also recommended to avoid sampling immediately after storms or heavy winds —at least 10–14 days after a major weather event (Hanke & Ruiz-Orejón, n.d.).

To minimize contamination during sample collection, researchers collecting plastics from seawater are advised to wear Personal Protective Equipment (PPE), including gloves, boots, and masks, to prevent personal exposure and sample contamination (Kannan et al., 2023). Each sample should be placed in clean, labeled stainless containers immediately after collection to avoid plastic contamination (Alvarado-Zambrano et al., 2024). The researchers must carefully avoid disturbing sediments or marine organisms during debris retrieval (S Lippiatt, 2020). Additionally, collecting macroplastics, researchers walked along transect lines during low tide to collect plastics from the shoreline and shallow areas (Acot et al., 2022; Pacilan & Bacosa, 2022; Sajorne et al., 2021). For deeper or less accessible areas, boat transects were conducted during high tide to ensure all parts of the area were covered (Dalu et al., 2025).

**Line Transect**

This study will use the line transect at low tide as one of its independent variables (Acot et al., 2022; Pacilan & Bacosa, 2022; Sajorne et al., 2021). According to Ryan (2022), this approach entails orderly gathering visible macroplastic waste along a predetermined coastal line during low

tide. When sampling at low tide, a larger intertidal region is exposed, facilitating the identification and recovery of stranded plastics larger than 5 mm (Karthik et al., 2021). Typically, each transect is 100 meters long and 5 to 10 meters wide on either side, guaranteeing uniform sampling coverage and site comparability (Acot et al., 2022; Pacilan & Bacosa, 2022). The line transect approach has been widely used in coastal monitoring programs to ensure consistent and repeatable data collection (Gaboy et al., 2022; Galgani et al., 2021).

**Boat Transect**

This study will use the boat transect at high tide as an independent variable (Dalu et al., 2025). With this approach, a boat travels pre-planned transect routes to gather floating or submerged macroplastic waste from deeper or less accessible coastal locations. By sampling at high tide, researchers can access offshore areas that are hidden at low tide, ensuring thorough coverage of the marine ecosystem. According to the Mekong River Commission (2023), each boat transect usually travels a straight line parallel to the beach, with standardized separations between sample locations to ensure uniformity. According to this method, the use of a net is crucial to capture the floating macroplastic waste, which is 0.1 mm - 0.3 mm (Mekong River Commission, 2023). This transect reduces bias and improves the representativeness of the data gathered under different tidal circumstances. Boat-assisted transects are very useful when evaluating the quantity, makeup, and distribution of macroplastics in open and semi-enclosed water environments (Galgani et al., 2021).

**Sample Categorization**

According to the Department of Environment and Natural Resources (DENR), samples must be weighed and then stored in clean, labeled polyethylene bags or stainless containers right after collection (PLASTIC POLICIES IN THE PHILIPPINES Country Profile 2, 2025). Each item will be

cleaned of remaining sediments or organic material using Hyclin Plus, followed by an initial rinse with tap water (Gao et al., 2023; Jones et al., 2024; Villegas-Camacho et al., 2024a). The researchers then used Milli-Q-grade distilled water to avoid interference during classification and FTIR analysis (Constant et al., n.d.; Villegas-Camacho et al., 2024b). Before packaging, plastics should be kept at room temperature for four hours with no air movement. This is necessary to prevent collected plastics from contaminating one another and to dry out any moisture (Villegas-Camacho et al., 2024a). Once dried, samples are sorted and categorized to help analyze their type, size, source, and degradation state (Galgani et al., 2021). The collected macroplastics will be weighed per plastic category, ranging from 0.11 g - 4.70 g or from 0.068 mg - 0.459 mg\cm2 (Gerritse et al., 2020a), depending on the density of each plastic. The size of the plastic sample will be carefully cut, no larger than 5 mm, using a Vernier caliper to maintain the same size in each cut sample (Villegas-Camacho et al., 2024b). The plastics then classified based on a modified framework from the study by Sajorne et al. (2021) covering everyday items such as (a) food packaging, (b) disposable utensils, (c) food containers, (d) cloth, (e) plastic caps, (f) sanitary napkins and diapers, (g) ropes and nylon, (h) cigarette butts, (i) plastic fragments, (j) sando bags, (k) styrofoam, (l) medicinal packaging, (m) sacks, (n) plastic bottles, (o) cellophane, and (p) other plastics.

**Sample Packaging**

In order to prevent plastic contamination, all materials utilized throughout the sampling, treatment, and characterization processes were composed of metal or glass (Alvarado-Zambrano et al., 2024). In an ecological field investigation, such as incorporating microplastic and macroplastic sediment collection, researchers preferably utilized non-plastic containers, such as stainless-steel containers, glass, and various metal instruments for gathering and moving samples

to the laboratory, since these containers reduce unwanted plastic contamination from plastic sampling equipment and storage (Iizuka et al., 2025; Nayebi et al., 2023).

**Sample Transportation**

Immediate and careful handling is essential to maintain the integrity of collected macroplastic samples. Proper transportation of macroplastic samples is necessary to prevent contamination, photodegradation, and moisture-related changes (Wang et al., 2024). Samples will be placed in clean, labeled polyethylene bags or containers immediately after collection (Koelmans et al., 2017). During transportation, samples will be sealed and protected from ultraviolet (UV) light, heat, rain, and physical contact to maintain their physical and chemical integrity (Koelmans et al., 2017).

**Polymer Analysis Using FTIR Spectroscopy**

At MSU-IIT Iligan, Fourier Transform Infrared (FTIR) spectroscopy will be used to identify the polymer composition of macroplastic samples (Arcadio et al., 2022; Villegas-Camacho et al., 2024). This technique produces an infrared spectrum for each sample. Also, the technique has identification aided by comparison with reference spectra (Kroon et al., 2018). Before analysis, researchers will clean and air-dry the samples to remove any debris or moisture that could change spectral results (Dalu et al., 2025; Nayebi et al., 2023). FTIR analysis identified polymers such as polyethylene (PE), polypropylene (PP), polystyrene (PS), and polyethylene terephthalate (PET).

## 1.3 Objectives of the Study

The main goal of this research is to analyze the macroplastic litter, particularly the source attribution of these plastics into microplastics in the seawater of Barangay Camagong, Nasipit, Agusan del Norte. The particular objectives of the study are as follows:

1. to determine the most common type of macroplastic per sampling site in the study

area;

2. to identify and categorize macroplastics based on form and material composition within the study area;
3. to determine the sources of polymer type of the collected macroplastic waste;
4. to relate the polymer type of the macroplastic sample to the microplastic presence in specific seafood; and
5. to provide evidence-based recommendations for local stakeholders based on the study findings.

**1.4 Scope and Limitations**

This study will focus on identifying the sources of microplastics in seawater. The researchers will conduct the study in the coastal waters of Barangay Camagong, Nasipit, Agusan del Norte, selected for their proximity to potential microplastic sources. The categories of plastics that the researchers will collect are only those greater than 5mm in size, visibly seen along mangrove, residential, and non-residential sites (Pacilan & Bacosa, 2022). The collected plastics will be categorized for polymer test analysis based on the three most prevalent items that will be found in the study area.

The plastic samples will be delivered to Mindanao State University, Iligan City, for polymer testing using Fourier Transform Infrared Spectroscopy (FTIR) micro-spectroscopy analysis. This technique confirmed the identification and classification of microplastics in oysters, as reported by Lumasag et al. (2024). However, this technique has limitations for identifying every particle, particularly those that are very small or have compositions similar to those of naturally occurring substances.

**1.5 Definition of Terms**

To further understand the contents of this research, the following are the keywords indicated in this study:

**Attribution**. Refers to tracing the source and possible contaminants of microplastics in Barangay Camagong, Nasipit, Agusan del Norte in seawater.

**Boat transect**. Refers to the method using a vessel to follow a predetermined path of collecting plastics in a body of water in Barangay Camagong, Nasipit, Agusan del Norte, during high tide.

**FTIR**. Stands for Fourier Transform Infrared spectroscopy, which can identify the type of polymer in plastic samples.

**Line transect**. Refers to the method of monitoring macroplastic debris along a long, predetermined straight shoreline during low tide in Barangay Camagong, Nasipit, Agusan del Norte.

**Mangrove site**. This sampling site refers to the vast mangrove forest near and on the shoreline of Barangay Camagong, Nasipit, Agusan del Norte.

**Macroplastic**. Refers to plastic items larger than 5 millimeters (0.5 cm) and visible during high and low tides, which the researchers will collect from seawater in Barangay Camagong, Nasipit, Agusan del Norte.

**Microplastic**. Refers to tiny plastic particles, typically less than 5 millimeters in size and not visible in naked eye, that accumulate in marine animals.

**Non-residential site**. This sampling site refers to the absence of any housing in Barangay Camagong, Nasipit, Agusan del Norte, in the seawater.

**Polymer Type**. Refers to the classification of the sources, structures, and properties of the plastic materials found in Barangay Camagong, Nasipit, Agusan del Norte.

**Residential site**. This sampling site refers to the community living near the coastal area in Barangay Camagong, Nasipit, Agusan del Norte, in the seawater.

**Seabed**. Refers to the ocean floor, where plastic debris settles and becomes visible during low tide.

**Seawater**. Refers to the water from Barangay Camagong, Nasipit, Agusan del Norte, which the researchers will use for macroplastic collection.

# CHAPTER 2

# RESEARCH METHODOLOGY

This chapter discusses the methodology to be used in this study, including the experimental research design, research materials and methods, research environment, population, sample, and ethical considerations. This study will focus on the collection of macroplastics from Barangay Camagong seawater to confirm the plastic polymer type that contributes to the bioaccumulation of microplastics in oysters (Villaceran et al., n.d.; Wootton et al., 2022) and in the same seawater of Barangay Camagong, as observed from the study of Lumasag et al. (2024). By following procedures and ethical practices, the research will ensure accurate results and provide insights into the sources of microplastics.

## 2.1 Research design

This study is experimental research that aims to identify the sources of microplastics by analyzing the polymer types of the most common macroplastics present in seawater. Experimental research will be used to collect data in a controlled setting to ensure precise outcomes. The researchers will gather the data by collecting, categorizing, and analyzing plastic materials from three selected sampling sites in the seawater of Barangay Camagong, Nasipit, Agusan del Norte. To further analyze the composition of plastic samples, Fourier Transform Infrared (FTIR) Spectroscopy will be used to determine their polymer types. This analysis will happen at the Centers for Sustainable Polymers (CSP) Laboratory at Mindanao State University, Iligan Institute of Technology (MSU-IIT).

The researchers will complete the method to secure approval from the appropriate authorities, including the Municipal Environment and Natural Resources (MENRO), Department of Environment and Natural Resources-City Environment and Natural Resources Office (DENR-

CENRO), the Local Government Unit of Nasipit, and the Barangay officials of Camagong, Nasipit, Agusan del Norte. The researchers will collaborate to ensure proper plastic collection at each site. Being sure about the collection involves determining the specific area size from the transect. The researchers will also seek approval from MSU-IIT for polymer testing of the samples to be collected. The researchers will submit the permission requests before the sample collection date. The researchers will send the plastic samples to MSU-IIT for polymer testing, where established protocols guide them to ensure accurate and reliable results. By comparing the polymer types found in macroplastic samples to those commonly identified in microplastic pollution, the study aims to establish a link between macroplastic waste and the presence of microplastics in seawater organisms. Through this method, the research aims to attribute microplastic contamination to specific sources, providing a scientific basis for targeted interventions in marine pollution management.

**Materials**

The materials needed for the experiment are as follows:

**Disposable gloves.** The researchers will use this to collect plastic samples from the three sites in Barangay Camagong sea to avoid contamination.



**Figure 1.** Disposable gloves

**Face masks.** The researchers will use this during plastic collection to ensure safety and prevent inhalation of microplastic particles.




**Figure 2.** Face masks

**Trash bags.** The researchers will use this to collect and properly categorize plastics from the seawater in Barangay Camagong.



**Figure 3.** Trash bags

**Trash bin.** The researchers will use this to store and contain the collected plastics during the collection process.



**Figure 4.** Trash bin

**Stainless steel ruler.** The researchers will use this to measure the length of macroplastics accurately.



**Figure 5.** Stainless steel ruler

**Analytical balance.** The researchers will use this to measure the weight of the macroplastics being collected.



**Figure 6.** Weighing scale

**Tongs.** The researchers will use this to sort plastics after manual collection.



**Figure 7.** Tong

**Fourier Transform Infrared Spectroscopy (FTIR).** It is an analytical technique that the researchers will use to identify and verify the polymer type of plastic in plastic samples.

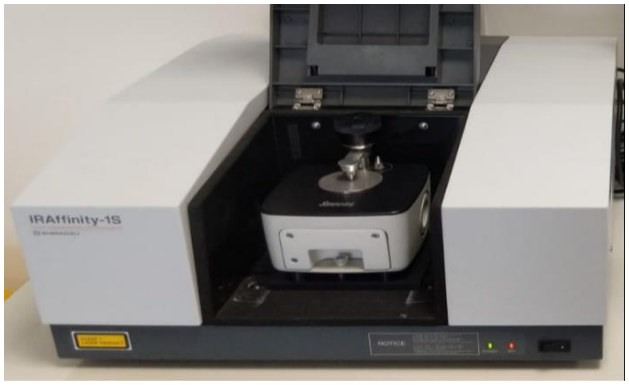



**Figure 8.** Fourier Transform Infrared Spectroscopy

**Steel tape.** The researchers will use this as a tool to measure the line transect during plastic sample collection.



**Figure 9.** Steel tape

**Waterproof boots.** The researchers will use this to protect feet from sharp objects, mud, and debris in seawater during plastic collection in the low tide season.



**Figure 10.** Waterproof boots

**Wooden stakes.** The researchers will use this to mark the start and end of each transect.



**Figure 11.** Wooden stakes

**Global Positioning System or GPs.** The researchers will use this to record the transect position, allowing for the repetition of sampling during subsequent collection sessions.




**Figure 12.** Global Positioning System

**Boat.** The researchers will use this only during high tide to access floating plastics in the seawater of Barangay Camagong.



**Figure 13.** Boat

**Oar.** The researchers will use this to manually navigate the boat during the collecting operation in the seawater of Barangay Camagong.



**Figure 14.** Oar

**Fishing net.** The researchers will use this to collect floating plastics and debris found on the water surface during the plastic collection operation.



**Figure 15.** Fishing net

**Stainless steel container.** The researchers will use this to store dried plastic to prevent contamination.



**Figure 16.** Stainless steel container

**Polyethylene storage container.** The researchers will use this container to store plastic sample for transportation to avoid prolonged exposure to the environment.



**Figure 17.** Polyethylene storage containers

**Vernier Caliper.** The researchers will use this to measure plastic in its appropriate size for polymer testing.



**Figure 18.** Vernier Caliper

**Milli-Q Distilled Water.** The researchers will use this to rinse plastics and remove surface contaminants, salts, and residues.




**Figure 19**. Milli-Q Distilled Water

**Hyclin Plus.** The researchers will use this as a disinfectant and cleaning solution to sanitize and remove organic contaminants, biofilms, or microbial residues from the plastic surface.




**Figure 20.** Hyclin Plus

**Scissor.** The researchers will use this to cut plastic in its appropriate size for polymer testing.



**Figure 21.** Scissor

**Disposable Surgical Gown.** The researchers will use this to minimize the harm and to prevent contamination, both to the plastics being collected and to the person collecting plastic.



**Figure 22.** Disposable Surgical Gown

**Safety Goggle.** The researchers will use this to protect the eyes from contact with contaminated water and physical debris.



**Figure 23.** Safety Goggle

**Face Shield**. The researchers will use this as a physical barrier to help protect the face and associated mucous membranes.



**Figure 24.** Face Shield

**Process flow**

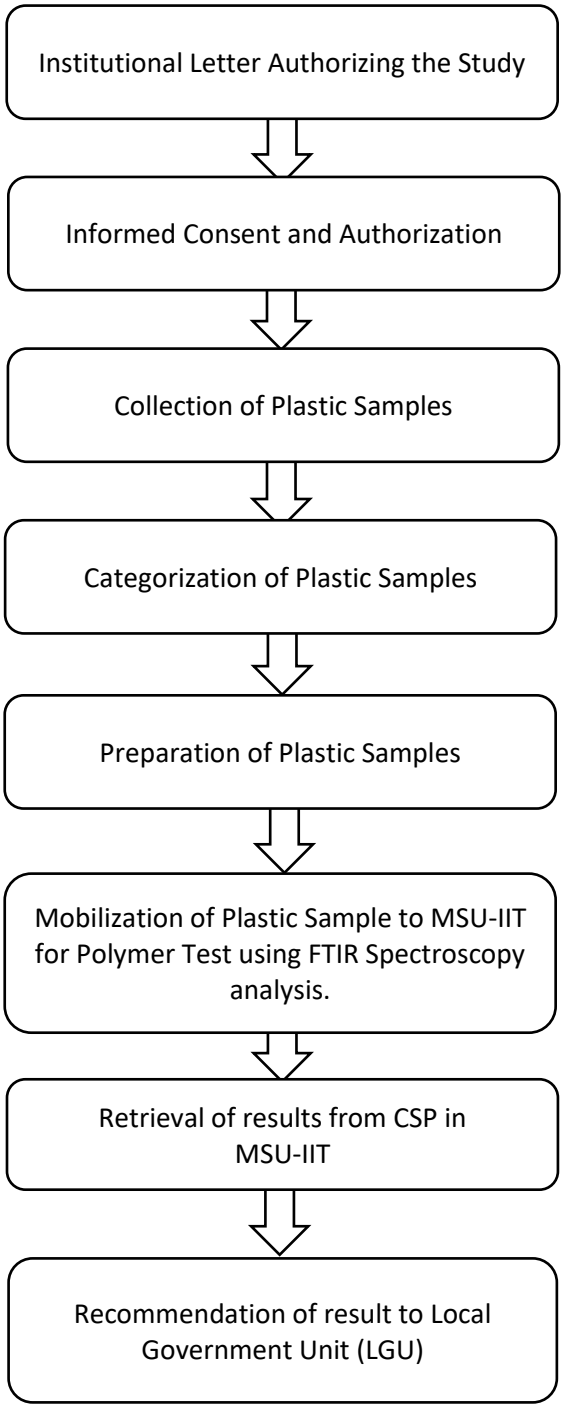



**Figure 25.** The diagram illustrates the process flow of the study, from obtaining authorizations to sharing findings.

**Methods**

The following procedures outline the collection of plastic samples from seawater at three designated sites in Barangay Camagong, Nasipit, Agusan del Norte.

**Collection of Plastics**

1. On the collection day, the researchers will gather at District 1, Barangay Camagong, Nasipit, Agusan del Norte to disseminate materials and PPE for collecting plastics during low tide, such as a trash bin, a trash bag, gloves, masks, surgical gowns, goggle, face shields, and boots, and during high tide, researchers will rent a boat with an oar and use fishing net for collecting plastics on the surface of the seawater.

2. The researchers will be collecting plastics in residential, non-residential, and mangrove sites for three weekends.

3. The researchers will wear Personal Protective Equipment (PPE) to collect plastics in the Barangay Camagong plastic samples.

4. The researchers will weigh the collected plastics.

5. The researchers will rinse the collected plastics with Hyclin plus, tap water, and Milli-Q distilled water, and then pat dry using a clean cloth and keep the plastics at room temperature without air currents.

6. Then, the dried plastics will be weighed.

**Categorization of Plastics**

1. After the drying and weighing process, the researchers will sort the collected plastics using tongs to determine which plastic category they belong to, and which among the plastic items are the most common in the study area.

2. The samples will then be weighed according to their respective categories.

**Preparation of Plastics**

1. The researchers will carefully cut the plastics into pieces that are not more than 5 mm in size to replicate the exact properties of microplastics.

2. Plastic samples will be stored in a clean stainless-steel container and placed in the polyethylene storage container.

3. Plastic samples will then be stored in a dust-free environment not exceeding room temperature, ready for transportation to The Center for Sustainable Polymers (CSP) laboratory in Bldg. 23, Mindanao State University - Iligan Institute of Technology, Andres Bonifacio Avenue, Tibanga, 9200 Iligan City, and used FT-IR, or Fourier Transform-Infrared Spectroscopy, for polymer test analysis, which will take approximately 5-6 hours.

**2.3 Research environment, population, and sample**

The study area will be in the seawater of Barangay Camagong, Nasipit, Agusan del Norte. The study site will be located in a coastal area inhabited by residents and mangroves, around 8°58' North latitude and 125°20' East longitude, in the northern region of Mindanao, within the province of Agusan del Norte. The researchers will set up one station with three sampling sites in District 1, Barangay Camagong. Their locations will be determined using Google Maps. Site 1 (8°58'27.1"N 125°20'15.6"E) will be the residential site of the NARWHAL and the mangrove site. The collection of plastic at this site will be during low tide. On the other hand, during high tide, collection of plastic is at Site 2 (8°58'30.0"N 125°20'13.4"E), where it is the coastal community and most of the fisherfolk resided, and Site 3 (8°58'30.2"N 125°20'09.2"E), which is the non-residential site, but near the Nasipit Port, where the active industrial shipping is located. According to the study by Lumasag et al. (2024), they selected these sites in Barangay Camagong because of potential environmental sources of microplastic pollution, particularly the evidence of macroplastics from

human activities, such as macroplastic discharges from the port due to commercial activities and residential communities, which contribute to the presence of microparticles in the seawater. Moreover, the study focuses on an area that includes Narwhal Park in Barangay Camagong, a coastal bay with mangrove forests and residential areas. This park also serves as the boundary between Barangay Camagong and its neighboring barangay, Barangay Talisay. The residents living within Narwhal Park are officially counted as part of Barangay Talisay. According to data from the Barangay Health Worker (BHW) of Barangay Talisay for the 4th quarter of 2025, Barangay Talisay comprises 166 households, distributed between District Molave (121 households) comprising 148 families and District Aurey (45 households) comprising 56 families, which are likely sources of plastic pollution.

For this study, the population comprises all macroplastic debris in the seawater under investigation that is larger than 5 millimeters and capable of undergoing physical, chemical, and biological degradation to form microplastic particles less than 5 millimeters (Yousafzai et al., 2025). These plastics serve as the secondary sources of macroplastics that the researchers will collect and sort into categories, which the researchers modified from the study of Sajorne et al. (2021), which include everyday items such as (a) food packaging, (b) disposable utensils (c) food containers, (d) cloth, (e) plastic caps, (f) sanitary napkin and diapers, (g) ropes and nylon, (h) cigarette butt, (i) plastic fragments, (j) sando bags, (k) styrofoam, (l) medicinal packaging, (m) sack, (n) plastic bottle, (o) cellophane and (p) other plastics of microplastics in seawater, and are thus central to source attribution (Borah et al., 2023; Hurley et al., 2025). The researchers will weigh first, then rinse the collected plastic with Hyclin plus, tap water, and Milli-Q distilled water (Gao et al., 2023; Jones et al., 2024; Villegas-Camacho et al., 2024a). The plastics are then patted dry with a clean cloth (Villegas-Camacho et al., 2024b). After drying for four hours in a room with no

airflow (Villegas-Camacho et al., 2024a) and weighing the dried samples (Gerritse et al., 2020b), the researchers will sort the plastics into categories based on the three most prevalent plastic items in the study area, then carefully cut them into pieces no larger than 5 mm to retain their microplastic properties. The researchers will use a Vernier caliper to cut the collected plastics to ensure all plastic samples are the same size. The goal is to create semi-square or rectangular pieces, depending on the material limitations (Villegas-Camacho et al., 2024b). Next, plastic sample classification will be based on nine common plastic polymers: polyethylene terephthalate (PET), high-density polyethylene (HDPE), polyvinyl chloride (PVC), low-density polyethylene (LDPE), polypropylene (PP), polystyrene (PS), polyamide (PA), multilayers, and others (Pacilan & Bacosa, 2022).

In this study, Barangay Camagong seawater will be selected for three sampling sites to allow researchers to collect macroplastics that may contribute to microplastic pollution through environmental degradation. These sites will be strategically selected based on proximity to potential pollution sources, including residential, non-residential, and mangrove areas. The collected macroplastics will be categorized by type, size, and the three most common items to help trace their possible origins and confirm their potential to fragment into microplastics over time.

**2.5 Research Ethical Standards**

**A. Intellectual Property Rights (IPR)**

Intellectual property rights will be upheld in this study, ensuring that all materials to create, including tools for data gathering, techniques, and analytical frameworks, will be protected under Philippine copyright law. The study will not involve the development of new technologies, products, or inventions that require patent protection. However, all original outputs, such as data

collection tools, methods, graphs, and reports, will be protected under copyright as the intellectual property of the researchers and Saint Michael College of Caraga. Any future use of the study materials will include proper acknowledgment of the authors.

**B. Informed Consent**

Before data collection, the researchers will seek permission from the Barangay Captain of Camagong, the Department of Environment and Natural Resources-City Environment and Natural Resources Office (DENR-CENRO), the Local Government Unit (LGU), and the Municipal Environment and Natural Resources Office (MENRO) of Nasipit. The informed consent for the boat driver during the boat transect and seek permission to use their time, which the researchers will compensate them for the time and effort to participate in the collection of plastics. The researchers will inform the residents of the purpose of the study and the activities to be conducted along the shoreline. The researchers will only include areas with local authority consent in the sampling. No human participants will be directly tested or subjected to experimentation in this research.

**C. Data Privacy and Confidentiality**

No personal information from residents will be included in this study. All data related to sampling locations, plastic types, and collected quantities will be stored securely and used solely for academic purposes. Only authorized members of the research team will have access to the data.

**D. Voluntary Participation**

Participation and cooperation from local authorities and community members will be voluntary. They may withdraw their consent at any time without penalty. The researchers will not provide monetary incentives or gifts to ensure impartiality and fairness.

**E. Minimization of Harm**

The researchers will collect macroplastics responsibly to prevent harm to marine life and coastal ecosystems. Researchers will wear Personal Protective Equipment (PPE), such as gloves, masks, surgical gowns, goggles, face shields, and boots, to maintain safety and prevent contamination. After analysis, all collected plastic waste will be properly segregated and disposed of in accordance with the Municipal Solid Waste Management Guidelines.

**F. Beneficence**

The research will aim to contribute positively to the coastal community of Barangay Camagong, Nasipit, Agusan del Norte, by providing essential information about the presence, types, and sources of macroplastic pollution along its shoreline. The results of this study will have potential long-term ecological and public health benefits for locals by supporting regional environmental initiatives and providing guidance for improving coastal waste management practices. The Department of Environment and Natural Resources Community Environment and Natural Resources Office (DENR-CENRO) Nasipit and the Local Government Unit (LGU) of Nasipit may utilize the findings to enhance solid waste management systems and reduce plastic leakage into marine environments.

**G. Justice and Fair Participant Selection**

Site selection will be based solely on environmental relevance, not socioeconomic or demographic characteristics. The researchers will ensure fair and unbiased treatment of all individuals involved during community coordination, upholding inclusivity and equal representation.

**H. Data Integrity and Accuracy**

Standardized scientific procedures for collecting, sorting, and measuring macroplastics will be followed to ensure reliability and validity. Data entries will be double-checked and verified for consistency. All possible biases or limitations encountered during fieldwork will be recorded and discussed transparently in the report.

**I. Transparency in Reporting**

The results of the study will be presented truthfully and openly. Researchers will report both positive and negative outcomes and remain committed to presenting results honestly and with integrity, including the sponsorships and donations offered to the researchers to help cover expenses during sample testing. The researchers will cite all references and data sources and use AI correctly to maintain transparency and protect intellectual property.

**J. Use of Patented or Copyrighted Materials**

The researchers will not use any patented or copyrighted materials without permission. Any literature, figures, or tools borrowed from other works will be cited appropriately. The researchers will respect all third parties' intellectual property rights throughout the study.

**K. Ethical Considerations for Human and Environmental Interaction**

No human or animal subjects will be used in experimental procedures. Environmental ethics will be observed by ensuring that no harm is caused to the natural surroundings during sampling. The research team will follow local and institutional guidelines for responsible ecological research practices.

**L. Responsible Use of AI and Related Technologies**

Artificial Intelligence (AI) tools, such as ChatGPT and Cici, will be used only for grammar checking, formatting, and organization of written output. These tools will also help summarize

related literature and suggest appropriate scientific wording, thereby improving the coherence and readability of the study. The researchers will refrain from using AI to generate research data or manipulate results. The researchers will ensure that the study's integrity, originality, and authorship are maintained at all times. Moreover, the researchers will use Fourier Transform Infrared (FTIR) Spectroscopy at the MSU-IIT CSP laboratory to determine the polymer types of the macroplastic samples to be collected. At MSU-IIT Iligan, Fourier Transform Infrared (FTIR) spectroscopy for polymer analysis identified polymers such as polyethylene (PE), polypropylene (PP), polystyrene (PS), and polyethylene terephthalate (PET).

**M. Ethical Approval**

The research proposal will be submitted to the Research Ethics Committee of Saint Michael College of Caraga for review and approval before fieldwork. The study will comply with institutional and national ethical standards governing environmental and scientific research.

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# APPENDICES

## A-1. LETTER OF APPROVAL

**Approval letter from the Department of Environment and Natural Resources- City Environment and Natural Resources Office (DENR-CENRO) Nasipit**

Saint Michael College of Caraga
Brgy. 4, Nasipit, Agusan del Norte, Philippines
District 8, Brgy. Triangulo, Nasipit, Agusan del Norte, Philippines
Tel. Nos. +63 085 343-3251 / +63 085 283-3113
www.smccnasipit.edu.ph

October 24, 2025

DENR-CENRO NASIPIT RECEIVED 28 OCT 2025 TIME 2:50 BY

**Nelson B. Caranzo**
**Chief Admin Officer**
DENR-CENRO Nasipit
Barangay Talisay, Nasipit, Agusan del Norte

Dear Sir Caranzo,

Greetings of peace and prosperity!

We are third-year BSED-Science students from Saint Michael College of Caraga. We are reaching out to respectfully request guidance and approval from your office for our research study, titled "Source Attribution of Microplastics in Seawater".This study is an expansion from the study of Lumasag et al. (2024) concluded that in Brgy. Camagong seawater, located at District 1 Brgy. Camagong, Nasipit, Agusan Del Norte, bioaccumulation of microplastics in oyster has the highest concentrations in residential site. Our research aims to investigate and analyze the macroplastic waste, specifically focusing on the source attribution of microplastics in seawater. As you may be aware, macroplastic pollution in seawater poses significant health risks to humans, particularly in relation to food safety and environmental sustainability, due to the degradation of plastics, which leads to microplastic contamination.

In line with this, we are asking for any environmental guidelines applicable in gathering macroplastic waste in the seawater during low tide and high tide for the study. We would greatly appreciate any guidelines your office can provide to help us conduct this research. We would also appreciate access to any existing data on macroplastic waste in the area that could support our study.

Please be assured that all information shared with us will be treated with strict confidentiality and used solely for academic purposes. If you have any questions or require additional information, please do not hesitate to contact us at 09535676207 or via email at jovieann_lumontad@smccnasipit.edu.ph

Thank you very much for considering our request.

Sincerely,

**JOVIE ANN B. LUMONTAD**
Research Leader

Noted by:

**DESIREE MAY H. ABIAN, RChE**
Research Adviser

Approved by:

**NELSON B. CARANZO**
Chief Admin Officer

**A-2. LETTER OF APPROVAL**

**Approval letter from the Barangay Camagong Office of Nasipit, Agusan del Norte**

Saint Michael College of Caraga
Brgy. 4, Nasipit, Agusan del Norte, Philippines
District 8, Brgy. Triangulo, Nasipit, Agusan del Norte, Philippines
Tel. Nos. +63 085 343-3251 / +63 085 283-3113
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October 24, 2025

**HON. RUDOLFO P. BULASA**
**Barangay Captain**
Barangay Camagong, Nasipit, Agusan del Norte

Dear Hon. Bulasa,

Greetings of peace and prosperity!

I hope this letter finds you in good health and high spirits. We are a dedicated team of student researchers from Saint Michael College of Caraga, and we are reaching out to request your permission to undertake a significant study in Barangay Camagong, Nasipit, Agusan del Norte. We are eager to collaborate with your esteemed leadership to explore a matter of great importance to our community.

Our research aims to investigate and analyze the macroplastic waste, specifically focusing on the source attribution of microplastics in seawater. As you may be aware, macroplastic pollution in seawater poses significant health risks to humans, particularly in relation to food safety and environmental sustainability, due to the degradation of plastics, which leads to microplastic contamination.

The results of our study will be shared with the Barangay Council and other relevant stakeholders to support any necessary actions that may be needed to mitigate the health risks associated with macroplastic pollution and microplastic contamination. Please be assured that our goal is not to create unnecessary concern but rather to provide valuable insights to help safeguard the health of the community.

We assure you that all research activities will be conducted in full compliance with ethical standards and local regulations. Furthermore, we will take every precaution to minimize any potential environmental and community impact throughout the research process.

We kindly request your support and permission to carry out this vital research. If you have any questions or require additional information, please do not hesitate to contact us at 09535676207 or via email at jovieann_lumontad@smccnasipit.edu.ph

Thank you very much for considering our request. We look forward to your positive response.

Sincerely,

**JOVIE ANN B. LUMONTAD**
Research Leader

Noted by:

**DESIREE MAY H. ABIAN, RChE**
Research Adviser

Approved by:

**HON. RUDOLFO P. BULASA**
Barangay Captain

## A-3. LETTER OF APPROVAL

**Approval letter from the Municipal Environment and Natural Resources of Nasipit, Agusan del Norte**

Saint Michael College of Caraga
Brgy. 4, Nasipit, Agusan del Norte, Philippines
District 8, Brgy. Triangulo, Nasipit, Agusan del Norte, Philippines
Tel. Nos. +63 085 343-3251 / +63 085 283-3113
www.smccnasipit.edu.ph

October 24, 2025

**HON. ROSCOE DEMOCRITO B. PLAZA**
**Municipal Mayor**
Nasipit, Agusan del Norte

OFFICE OF THE MUN. MAYOR
NASIPIT, AGUSAN DEL NORTE
RECEIVED
BY:
DATE: 10/30/25
TIME: 1:44 pm

Thru: **Michaele Joefrey B. Cassion**
**Head, MENRO**
Nasipit, Agusan del Norte

Dear Hon. Plaza,

Greetings of peace and prosperity!

We are third-year BSED-Science students from Saint Michael College of Caraga. We are reaching out to respectfully request guidance and approval from your office for our research study, titled "Source Attribution of Microplastics in Seawater".This study is an expansion from the study of Lumasag et al. (2024) concluded that in Brgy. Camagong seawater, located at District 1 Brgy. Camagong, Nasipit, Agusan Del Norte, bioaccumulation of microplastics in oyster has the highest concentrations in residential site. Our research aims to investigate and analyze the macroplastic waste, specifically focusing on the source attribution of these plastics in seawater. As you may be aware, macroplastic pollution in seawater poses significant health risks to humans, particularly in relation to food safety and environmental sustainability, due to the degradation of plastics, which leads to microplastic contamination.

In line with this, we are asking for any environmental guidelines applicable in gathering macroplastic waste in the seawater during low tide and high tide for the study. We would greatly appreciate any guidelines your office can provide to help us conduct this research. We would also appreciate access to any existing data on macroplastic waste in the area that could support our study.

Please be assured that all information shared with us will be treated with strict confidentiality and used solely for academic purposes. If you have any questions or require additional information, please do not hesitate to contact us at 09535676207 or via email at jovieann_lumontad@smccnasipit.edu.ph

Thank you very much for considering our request.

Sincerely,

**JOVIE ANN B. LUMONTAD**
Research Leader

Noted by:

**DESIREE MAY H. ABIAN, RChE**
Research Adviser

Approved by:

**HON. ROSCOE DEMOCRITO B. PLAZA**
Municipal Mayor

## A-4. LETTER OF APPROVAL

### Approval letter from the Local Government Unit of the Nasipit Municipal Office

Saint Michael College of Caraga
Brgy. 4, Nasipit, Agusan del Norte, Philippines
District 8, Brgy. Triangulo, Nasipit, Agusan del Norte, Philippines
Tel. Nos. +63 085 343-3251 / +63 085 283-3113
www.smccnasipit.edu.ph

October 24, 2025

**HON. ROSCOE DEMOCRITO B. PLAZA**
**Municipal Mayor**
Nasipit, Agusan del Norte

OFFICE OF THE MUN. MAYOR
NASIPIT, AGUSAN DEL NORTE
RECEIVED
BY:
DATE:
TIME: 1:44 pm

Dear Hon. Plaza,

Greetings of peace and prosperity!

We are a dedicated team of student researchers from Saint Michael College of Caraga, and we are reaching out to formally request permission to undertake our current study on the Source Attribution of Microplastics in Seawater at District 1, Barangay Camagong, Nasipit, Agusan del Norte.

The proposed research study aims to investigate and analyze the macroplastic waste, specifically focusing on the source attribution of microplastics in seawater. Our research team will collect information about macroplastic pollution in seawater that poses significant health risks to humans, particularly in relation to food safety and environmental sustainability, due to the degradation of plastics, which leads to microplastic contamination, which would be an environmental implication in our aquatic ecosystem.

We recognize the importance of obtaining proper authorization and support from the municipal government in order to conduct this study responsibly and ethically. We assure you that all research activities will adhere to relevant regulations and guidelines, and we are committed to minimizing any disruption to local aquaculture practices.

The results of our study will be shared with the Barangay Council and other relevant stakeholders to support any necessary actions that may be needed to mitigate the health risks associated with macroplastic pollution and microplastic contamination. Given the significance of this study, we kindly seek your approval and cooperation in facilitating our research.

We are enthusiastic about the opportunity to collaborate with our town's administration, and we are available to provide further information or address any inquiries you may have. If you have any questions or require additional information, please do not hesitate to contact us at 09535676207 or via email at jovieann_lumontad@smccnasipit.edu.ph

Thank you very much for considering our request, and your active engagement is the catalyst we need to profoundly advance scientific knowledge and champion crucial environmental stewardship, forging a sustainable and enlightened future for our town's researchers.

Sincerely,

**JOVIE ANN B. LUMONTAD**
Research Leader

Noted by:

**DESIREE MAY H. ABIAN, RChE**
Research Adviser

Approved by:

**HON. ROSCOE DEMOCRITO B. PLAZA**
Municipal Mayor

**B. INFORMED CONSENT**

SAINT MICHAEL COLLEGE OF CARAGA
Atupan St., Brgy. 4, Nasipit, Agusan del Norte 8602, Philippines
Website: www.smccnasipit.edu.ph ; Tel. Nos. 085 300-2932

**SMCC-REC FORM 4**
**RESEARCHER'S INFORMED CONSENT TO PARTICIPANTS**

| | |
|---|---|
| **TITLE OF THE STUDY**<br>SOURCE ATTRIBUTION OF MICROPLASTICS IN SEAWATER | |
| **NAME/S OF THE RESEARCHER/S:**<br>JOVIE ANN B. LUMONTAD<br>KRISTINE NICY I. ANOBLING<br>ANDY ANGELO B. CADAVOS<br>MARIE ANTOINETTE E. CUSTODIO<br>JEE KYLE M. LOPEZ<br>JACKY JUN D. MINAO | |
| **DEPARTMENT:**<br>COLLEGE OF TEACHER EDUCATION | **COURSE:**<br>BACHELOR OF SECONDARY EDUCATION<br>MAJOR IN SCIENCE |
| **INTRODUCTION**<br>You are being invited to participate in a research study conducted by researcher/researchers above who are affiliated with Saint Michael College of Caraga.<br><br>You have been selected as a potential informant because you meet the criteria necessary for inclusion in our study.<br><br>Your participation in this study is entirely voluntary. We encourage you to take your time to review the information provided below. Please feel free to ask any questions or seek clarification on any part of the study that you do not fully understand before deciding whether to participate. Additionally, you may choose to discuss your potential involvement in this research with family members or friends.<br><br>Should you decide to participate, you will be asked to sign this consent form. A copy of this form will be provided for your records. | |
| In case of concerns about the study, you may contact the Research And Instructional Innovation Deparment through the following:<br>• Email: riid@smccnasipit.edu.ph<br>• Contact number: 09778064574<br><br>Or the researchers of this study:<br>• EMAIL: jovieann_lumontad@smccnasipit.edu.ph<br>• CONTACT NUMBER: 09535676207 | |
| **PURPOSE OF THE STUDY** | This study aims to accomplished the following:<br>1. Determine the predominant macroplastic types present at each sampling location within the designated study area.<br>2. Systematically identify and categorize macroplastics based on their form and material composition within the defined study area.<br>3. Identify the origins of the polymer types identified in the collected macroplastic waste samples.<br>4. Establish a correlation between the polymer types |

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Atupan St., Brgy. 4, Nasipit, Agusan del Norte 8602, Philippines
Website: www.smccnasipit.edu.ph ; Tel. Nos. 085 300-2932

| | |
|---|---|
| | found in macroplastic samples and the occurrence of microplastics in selected seafood species.<br>5. Generate evidence-based recommendations for relevant local stakeholders, informed by the comprehensive findings of this research endeavor. |
| **PROCEDURES OF THE STUDY** | Please Check below which type of study is being conducted<br>☐ For Quantitative and Survey Questionnaires:<br>*If you volunteer to participate in this study, you will be asked to participate by answering the survey questionnaire which you can finish in less than 30 minutes.*<br>☐ For Interviews/ FGD / Case Studies<br>*If you volunteer to participate in this study, you will be asked to participate by answering the questions in the interview guide which may last 30-40 minutes depending on the length of the interview. Moreover, this will be recorded through audio or video.*<br>☒ For capstones and experimental studies<br>*If you volunteer to participate in this study, you will be asked to participate by allowing the researchers to use specimens and other data needed related to the research.* |

**POTENTIAL RISKS AND DISCOMFORTS**

You may experience some discomfort during the interview/questionnaires due to the sensitive nature of the topics being discussed. If at any point you encounter questions that cause psychological or emotional distress, you have the right to decline to answer without any consequence. Additionally, you may choose to withdraw from the study at any time if you feel unable to continue discussing the requested information. The researchers prioritize your well-being and will ensure that your comfort and welfare remain their highest priority throughout the study.

**POTENTIAL BENEFITS TO PARTICIPANTS AND/OR TO SOCIETY**

This study aims to generate valuable insights that may benefit both public and private sector administrators, as well as human resource managers and policy-makers. The findings and discussions derived from this research can provide evidence-based information that may inform decision-making processes and strategic planning. Additionally, the results of this study may be of particular interest to government agencies such as:

1. The Barangay and its Residents. This study gives key information about the types and sources of plastic polluting its waters by helping to create better waste management programs and improve the overall health of the community and its environment.
2. The Local Government Unit (LGU). The data allows to create effective policies and waste management plans to protect seafood, promote sustainability, improve community well-being, and encourage collaboration between the LGU, schools, and community members to fight microplastic pollution.
3. Saint Michael College of Caraga (SMCC). The research findings can support community extension program initiated by SMCC, focusing on plastic waste management and promoting seawater quality, responsible waste disposal practices, and community awareness regarding the impact of plastic pollution on marine life.
4. The Broader Community. This study offers a blueprint for communities struggling with plastic

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Atupan St., Brgy. 4, Nasipit, Agusan del Norte 8602, Philippines
Website: www.smccnasipit.edu.ph ; Tel. Nos. 085 300-2932

pollution and microplastics in marine life that promotes sustainable practices and provides essential information for implementing proper waste disposal, ultimately reducing microplastic contamination in marine animals and protecting ecosystems.

**CONFIDENTIALITY**

All information gathered in this study will be treated with the utmost confidentiality, in accordance with applicable laws and regulations. Any identifiable information collected will be kept strictly confidential, except in cases where disclosure is necessary to protect your rights or welfare. This confidentiality agreement ensures that the researchers can protect your participation information from being accessed by individuals who are not directly involved in the study. When the results of this research are published or presented at conferences, your identity will not be revealed, and no personally identifiable information will be used.

**PARTICIPATION AND WITHDRAWAL**

Your participation in this study is entirely voluntary. Choosing not to participate will not result in any penalty or loss of benefits to which you are otherwise entitled. By participating in this study, you are not waiving any legal claims, rights, or remedies. You may withdraw from the study at any time without any negative consequences.

**RESEARCH PARTICIPANT'S CONSENT**

I have read and understood the information provided above. I have had the opportunity to ask questions, and my questions have been answered to my satisfaction. I understand that I will receive a copy of this consent form. I am aware that I can withdraw my consent at any time and discontinue my participation in the study without any penalties or loss of benefits to which I am otherwise entitled.

Name And Signature of The Participant

Date Signed:

***IF ILLITERATE***

I have witnessed the accurate reading of the consent form to the potential participant, and the individual has had the opportunity to ask questions. I confirm that the individual has given consent freely.

Name And Signature Of The Partcipant

Thumb Print

**RESEARCHER'S CERTIFICATION**

**B-1. INFORMED CONSENT**

**Barangay Captain of Brgy. Camagong, Nasipit, Agusan del Norte**

SAINT MICHAEL COLLEGE OF CARAGA
Atupan St., Brgy. 4, Nasipit, Agusan del Norte 8602, Philippines
Website: www.smccnasipit.edu.ph ; Tel. Nos. 085 300-2932

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Name And Signature of The Participant

Date Signed:

***IF ILLITERATE***

I have witnessed the accurate reading of the consent form to the potential participant, and the individual has had the opportunity to ask questions. I confirm that the individual has given consent freely.

Name And Signature Of The Partcipant

Thumb Print

**RESEARCHER'S CERTIFICATION**

I have explained the details of the research study to the participant, answered all questions posed, and ensured that the participant fully understands the information provided in this document. I believe that the participant has freely given their consent to participate in the study.

JOVIE ANN B. WIMONTAO
NAME AND SIGNATURE OF THE RESEARCHER
Date Signed: 01-23-26

**Confirmation of Participant/s**

RODOLFO P. BULASA
NAME AND SIGNATURE OF THE PARTICIPANT
Date Signed: 01-26-26

**B-2. INFORMED CONSENT**

**Department of Environment and Natural Resources-City Environment and Natural Resources Office (DENR-CENRO)**

SAINT MICHAEL COLLEGE OF CARAGA
Atupan St., Brgy. 4, Nasipit, Agusan del Norte 8602, Philippines
Website: www.smccnasipit.edu.ph ; Tel. Nos. 085 300-2932

pollution and microplastics in marine life that promotes sustainable practices and provides essential information for implementing proper waste disposal, ultimately reducing microplastic contamination in marine animals and protecting ecosystems.

**CONFIDENTIALITY**

All information gathered in this study will be treated with the utmost confidentiality, in accordance with applicable laws and regulations. Any identifiable information collected will be kept strictly confidential, except in cases where disclosure is necessary to protect your rights or welfare. This confidentiality agreement ensures that the researchers can protect your participation information from being accessed by individuals who are not directly involved in the study. When the results of this research are published or presented at conferences, your identity will not be revealed, and no personally identifiable information will be used.

**PARTICIPATION AND WITHDRAWAL**

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**RESEARCH PARTICIPANT'S CONSENT**

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Name And Signature of The Participant

Date Signed:

***IF ILLITERATE***

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Name And Signature Of The Partcipant

Thumb Print

**RESEARCHER'S CERTIFICATION**

I have explained the details of the research study to the participant, answered all questions posed, and ensured that the participant fully understands the information provided in this document. I believe that the participant has freely given their consent to participate in the study.

JOVIE ANN B. LUMONTAD
NAME AND SIGNATURE OF THE RESEARCHER
Date Signed: 01-23-26

**Confirmation of Participant/s**

MAICA CASAGDA
NAME AND SIGNATURE OF THE PARTICIPANT
Date Signed: 01/24/26

**B-3. INFORMED CONSENT**

**Local Government of Nasipit, Agusan del Norte**

SAINT MICHAEL COLLEGE OF CARAGA
Atupan St., Brgy. 4, Nasipit, Agusan del Norte 8602, Philippines
Website: www.smccnasipit.edu.ph ; Tel. Nos. 085 300-2932

**SMCC-REC FORM 4**
**RESEARCHER'S INFORMED CONSENT TO PARTICIPANTS**

| |
|---|
| **TITLE OF THE STUDY**<br>SOURCE ATTRIBUTION OF MICROPLASTICS IN SEAWATER |
| **NAME/S OF THE RESEARCHER/S:**<br>JOVIE ANN B. LUMONTAD — MARIE ANTOINETTE E. CUSTODIO<br>KRISTINE NICY I. ANOBLING — JEE KYLE M. LOPEZ<br>ANDY ANGELO B. CADAVOS — JACKY JUN D. MINAO |
| **DEPARTMENT:** COLLEGE OF TEACHER EDUCATION — **COURSE:** BACHELOR OF SECONDARY EDUCATION MAJOR IN SCIENCE |
| **INTRODUCTION**<br>You are being invited to participate in a research study conducted by researcher/researchers above who are affiliated with Saint Michael College of Caraga.<br><br>You have been selected as a potential informant because you meet the criteria necessary for inclusion in our study.<br><br>Your participation in this study is entirely voluntary. We encourage you to take your time to review the information provided below. Please feel free to ask any questions or seek clarification on any part of the study that you do not fully understand before deciding whether to participate. Additionally, you may choose to discuss your potential involvement in this research with family members or friends.<br><br>Should you decide to participate, you will be asked to sign this consent form. A copy of this form will be provided for your records. |
| In case of concerns about the study, you may contact the Research And Instructional Innovation Deparment through the following:<br>• Email: riid@smccnasipit.edu.ph<br>• Contact number: 09778064574<br><br>Or the researchers of this study:<br>• EMAIL: jovieann_lumontad@smccnasipit.edu.ph<br>• CONTACT NUMBER: 09535676207 |
| **PURPOSE OF THE STUDY** — This study aims to accomplished the following:<br>1. Determine the predominant macroplastic types present at each sampling location within the designated study area.<br>2. Systematically identify and categorize macroplastics based on their form and material composition within the defined study area.<br>3. Identify the origins of the polymer types identified in the collected macroplastic waste samples.<br>4. Establish a correlation between the polymer types |

OFFICE OF THE MUN. MAYOR
NASIPIT, AGUSAN DEL NORTE
RECEIVED
BY:
DATE: 11/26/24
TIME: 2:24 PM

Scanned with CamScanner

**B-4. INFORMED CONSENT**

**Boat Operator During Plastic Collection Under High Tide Condition**

**SAINT MICHAEL COLLEGE OF CARAGA**
Atupan St., Brgy. 4, Nasipit, Agusan del Norte 8602, Philippines
Website: www.smccnasipit.edu.ph ; Tel. Nos. 085 300-2932

pollution and microplastics in marine life that promotes sustainable practices and provides essential information for implementing proper waste disposal, ultimately reducing microplastic contamination in marine animals and protecting ecosystems.

**CONFIDENTIALITY**

All information gathered in this study will be treated with the utmost confidentiality, in accordance with applicable laws and regulations. Any identifiable information collected will be kept strictly confidential, except in cases where disclosure is necessary to protect your rights or welfare. This confidentiality agreement ensures that the researchers can protect your participation information from being accessed by individuals who are not directly involved in the study. When the results of this research are published or presented at conferences, your identity will not be revealed, and no personally identifiable information will be used.

**PARTICIPATION AND WITHDRAWAL**

Your participation in this study is entirely voluntary. Choosing not to participate will not result in any penalty or loss of benefits to which you are otherwise entitled. By participating in this study, you are not waiving any legal claims, rights, or remedies. You may withdraw from the study at any time without any negative consequences.

**RESEARCH PARTICIPANT'S CONSENT**

I have read and understood the information provided above. I have had the opportunity to ask questions, and my questions have been answered to my satisfaction. I understand that I will receive a copy of this consent form. I am aware that I can withdraw my consent at any time and discontinue my participation in the study without any penalties or loss of benefits to which I am otherwise entitled.

Name And Signature of The Participant

Date Signed:

***IF ILLITERATE***

I have witnessed the accurate reading of the consent form to the potential participant, and the individual has had the opportunity to ask questions. I confirm that the individual has given consent freely.

Name And Signature Of The Partcipant

Thumb Print

**RESEARCHER'S CERTIFICATION**

I have explained the details of the research study to the participant, answered all questions posed, and ensured that the participant fully understands the information provided in this document. I believe that the participant has freely given their consent to participate in the study.

JOVIE [illegible] D. CUNONTAD
NAME AND SIGNATURE OF THE RESEARCHER
Date Signed: 01-23-26

**Confirmation of Participant/s**

NAME AND SIGNATURE OF THE PARTICIPANT
Date Signed: 01/23/2026

## C. MAP OF THE RESEARCH LOCALE

